

ATAD2 mediates chromatin-bound histone chaperone turnover

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eLife Assessment

This **valuable** study explores the role of the chromatin regulator ATAD2 in mouse spermatogenesis. It **convincingly** demonstrates that ATAD2 is essential for proper chromatin remodeling in haploid spermatids, influencing gene accessibility, H3.3-mediated transcription, and histone eviction. Using *Atad2* knockout (KO) mice, the authors link ATAD2 to the DNA-replication-independent incorporation of sperm-specific proteins like protamines and histone H3.3. Although the findings highlight chromatin abnormalities and impaired in vitro fertilization in KO mice, natural fertility remains unaffected, suggesting possible in vivo compensatory mechanisms. However, in its current form, the study lacks mechanistic insight and provides only partial evidence for ATAD2's molecular role, limiting its functional conclusions.

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Abstract

ATAD2, a conserved protein which is predominantly expressed in embryonic stem (ES) cells and spermatogenic cells, emerges as a crucial regulator of chromatin dynamics. Our previous parallel studies in both ES cells and *S. pombe* highlighted the key role of ATAD2 in facilitating chromatin-bound histone chaperone turnover. Focusing on spermatogenesis, we demonstrate here that ATAD2 regulates the HIRA-dependent localization of H3.3 on the genome and influences H3.3-mediated gene transcription. Moreover, by modulating histone eviction and the assembly of protamines, ATAD2 ensures proper chromatin condensation and genome packaging in mature sperm. Disruption of *Atad2* function in mice leads to abnormal genome organization in mature spermatozoa. Together, these findings establish a previously overlooked level of chromatin dynamic regulation, governed by ATAD2's interaction with histone chaperones and chromatin, which controls the balance between histone deposition and removal.

Introduction

Spermatogenesis involves unique and dramatic chromatin remodelling and genome reorganization events that take place in preparation for a shift from the universal nucleosome-based eukaryotic genome organization into a unique nucleoprotamine-based structure (Gaucher et al. 2010 [↗](#); Bao and Bedford 2016 [↗](#); Le Blevec et al. 2020 [↗](#); Okada 2022 [↗](#)). During the last decade, important information on the molecular basis of the nucleosome-to-nucleoprotamine transition during the postmeiotic phases of spermatogenesis became available. Several testis-specific H2A, H2B and H3 histone variants replace the canonical histones, locally or globally, at different stages, starting early during the commitment of cells into meiotic and post-meiotic differentiation (Hoghoughi et al. 2018 [↗](#)). Most of this nucleosome dismantlement and assembly occurs in the absence of DNA replication. In other words, the replication-independent nucleosome assembly pathways, such as the well-documented HIRA-dependent H3.3 assembly, are expected to be critical in ensuring the necessary chromatin dynamics in these cells. Finally, hints to the functions of transition proteins (TPs), and their functional relationship with protamines (PRMs) were proposed based on the study of a specific late-expressing H2A variant, H2A.L.2 (Barral et al. 2017b [↗](#)). In the proposed model we suggest that TPs do not directly replace histones, but rather could be loaded on open H2A.L.2-containing nucleosomes to control PRM assembly. Nucleosome-to-nucleoprotamine transition also includes a TP-dependent pre-PRM2 processing, histone displacement by PRMs and the structuration of the nucleoprotamine-based genome organization in the mature spermatozoa (Barral et al. 2017b [↗](#); Rezaei-Gazik et al. 2022 [↗](#)).

Taking into account this unique dynamic of post-meiotic cells' nucleosomes, it is expected that these cells massively express generic or germline-specific factors that act on chromatin, including histone acetylating machineries, acetylated histone readers and histone variants and their chaperones, all of which mediate histone and non-histone protein exchanges. More specifically, the replication-independent nature of nucleosome assembly, both in cells undergoing meiosis and in post-meiotic cells, strongly suggests the involvement of the chaperone / histone substrate couple, HIRA and H3.3.

An important part of this replication-independent histone assembly and transitions in the genome organizations occurs in the context of a massive histone H4 hyperacetylation in the haploid spermatogenic cells, spermatids, mediated by NUT-p300/CBP histone acetyltransferase complex (Shiota et al. 2018 [↗](#); Rousseaux et al. 2022 [↗](#)). Hyperacetylation of histone H4, especially, the double acetylation on H4K5K8 in these cells, leads to the binding of H4 by the first bromodomain of BRDT and to histone replacement (Moriniere et al. 2009 [↗](#); Gaucher et al. 2012 [↗](#); Goudarzi et al. 2016 [↗](#); Shiota et al. 2018 [↗](#)). Additionally, Histone H4 hyperacetylation, especially at K5, could also be recognized by another bromodomain-containing factor of unknown function in spermatogenic cells, ATAD2 (Caron et al. 2010 [↗](#); Morozumi et al. 2016 [↗](#)).

ATAD2 is a highly conserved bromodomain-containing factor harbouring, in addition, an AAA ATPase domain (Boussouar et al. 2013 [↗](#); Cattaneo et al. 2014 [↗](#)). Our early work showed that Atad2 is predominantly expressed in embryonic stem cells (ES cells), as well as in spermatogenic cells, and is aberrantly activated in virtually all solid cancers (Caron et al. 2010 [↗](#)). Although ATAD2's function in cancer prompted many investigations (Liu et al. 2022 [↗](#)), its function in its physiological context of expression has remained largely overlooked. We reported the first investigations of ATAD2's function in embryonic stem cells (Morozumi et al. 2016 [↗](#)). These studies in ES cells showed that ATAD2 is recruited into acetylated transcriptionally active chromatin and is critical to keep the chromatin of ES cells highly dynamic, as illustrated by an ATAD2-dependent high rate of histone exchange (Morozumi et al. 2016 [↗](#)). Later parallel studies by our laboratory, in ES cells and in *S. pombe*, deciphered the molecular basis of this activity of ATAD2, which relies on its capacity to ensure the turnover of the chromatin-bound histone chaperone, HIRA and FACT

(Wang et al. 2021 [↗](#)). In addition to ES cells, *Atad2* is also highly expressed in male germ cells, pointing to its potential role in the regulation of HIRA -dependent H3.3 assembly in these cells, especially, in post-meiotic cells. Indeed, previous published works have highlighted the important function of H3.3 assembly in meiotic and post-meiotic male genome programming (van der Heijden et al. 2007 [↗](#); Yuen et al. 2014 [↗](#); Tang et al. 2015 [↗](#); Fontaine et al. 2022 [↗](#)). We therefore hypothesized that ATAD2-controlled HIRA-H3.3 activity would also be a critical player in post-meiotic male genome reorganization, as HIRA has been shown to be essential post-fertilization (Lin et al. 2014 [↗](#)) and during oogenesis (Nashun et al. 2015 [↗](#)).

Using a lacZ reporter *Atad2* KO mouse model, we could show that *Atad2*'s expression is dramatically enhanced in post-meiotic male germ cells, ensuring a proper dynamics of histone H3.3, including on the sex chromosomes. More remarkably, our data demonstrate that ATAD2 controls the appropriate expression of a series of genes whose activity is known to be H3.3-dependent in post-meiotic cells. Additionally, in the absence of ATAD2, perturbed nucleosome dynamics leads to a delayed histone-to-PRM-replacement, leading to a defective global final male genome compaction. Although this defective mature spermatozoon genome compaction has little effect on male fertility following natural reproduction, it impacts the success of *in vitro* fertilization using spermatozoa from *Atad2* KO mice pointing to a more fragile genome in these cells.

In summary, our work identified a yet unknown level of the regulation of chromatin plasticity, that involves the control of histone chaperone dynamics, particularly that of HIRA, controlling the proper male germ cell transcriptional regulation, genome compaction and optimal male fertility.

Results

Atad2 is predominantly expressed in haploid male germ cells

Atad2 is normally expressed in ES and spermatogenic cells and remains silent in the majority of adult tissues (Morozumi et al. 2016 [↗](#)). Therefore, in addition to ES cells, male germ cells constitute the second cell type where the molecular basis of the physiological ATAD2 activity could be investigated.

To decipher the function of ATAD2 during spermatogenesis, we used a mouse model harboring an insertion of a *lacZ* reporter cassette between exons 11 and 12 that enables fusion of the *lacZ* transgene with exon 11 of the long form of *Atad2* (Fig. 1A [↗](#)) and production of a non-functional ATAD2-lacZ fusion protein (Fig. 1B [↗](#)), referred to as ATAD2 KO in this study. Furthermore, we crossed these mice with transgenic mice expressing ubiquitously Cre recombinase under *CMV* promoter, in order to remove the *neo* cassette and to delete *Atad2*'s exon 12 (Fig. 1B [↗](#)). In addition to the long somatic form of ATAD2, ATAD2L, the mouse testis encodes a short germ-cell specific isoform that we previously named ATAD2S (Caron et al. 2010 [↗](#)). The combined LacZ fusion and *Cre recombinase*-dependent deletion of *Atad2*'s exon 12 make both ATAD2L and ATAD2S undetectable (Fig. 1C [↗](#)).

The expression of *lacZ* under the control of the *Atad2* promoter allows one to visualize *Atad2* gene activity in a stage-specific manner. X-gal staining of seminiferous tubule sections from *Atad2* KO mice showed a remarkably high expression of *Atad2* in post-meiotic spermatogenic cells (Fig. 1D [↗](#)). It is noteworthy that no staining was observed in the wild-type tubules.

ATAD2 controls HIRA accumulation

In our previous work performed in both *S. pombe* and mouse ES cells, we demonstrated the crucial role for ATAD2 in HIRA-dependent histone dosage as well as in HIRA-chromatin interaction. Among other discoveries, this work showed that in the absence of ATAD2, HIRA becomes trapped

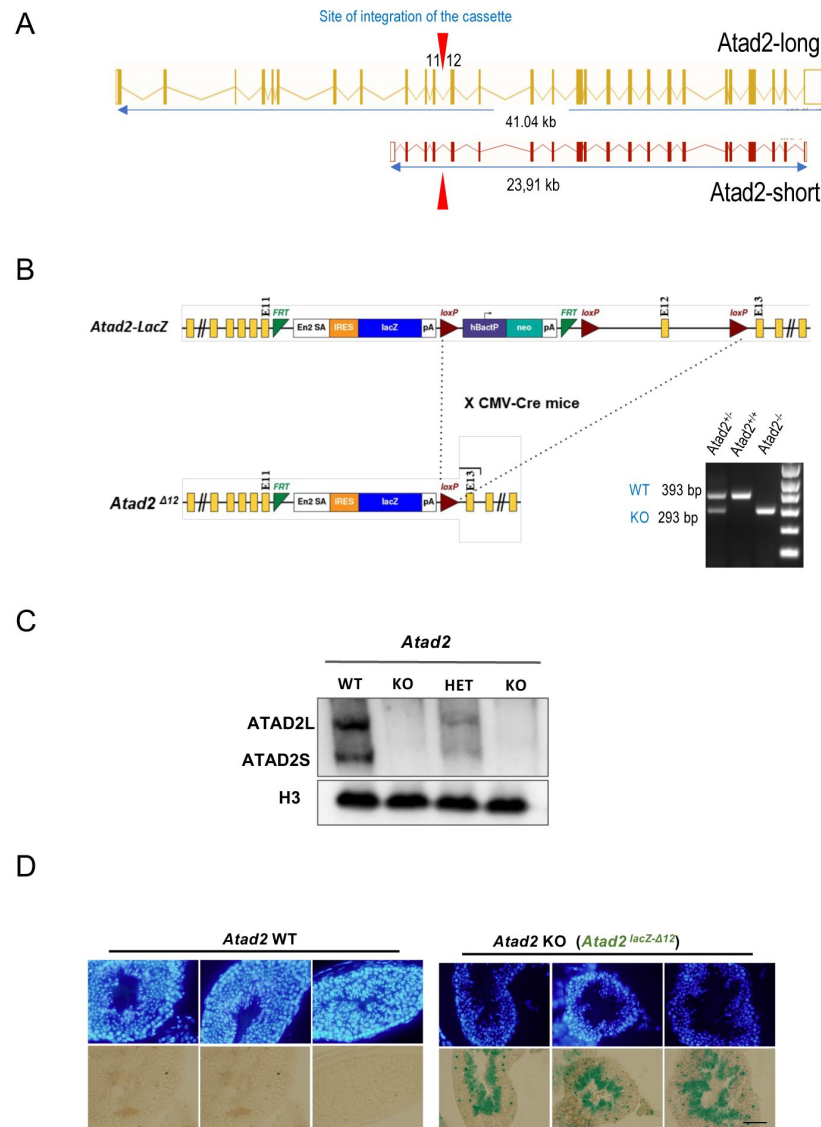


Figure 1

Atad2 is highly expressed in post-meiotic male germ cells.

(A) The exon/intron organization of *Atad2* gene and the two ATAD2-encoding transcriptional units (encoding for ATAD2-long and ATAD2-short) are indicated (Dyer et al. 2025). The red arrows indicate the position of insertion of the *LacZ-Lox-Neo-Lox* cassette. (B) Gene targeting strategy for the generation of the *Atad2* KO allele is represented. Exon number and relative position, selection gene cassettes, LoxP sites and Frt sites are indicated. Crossing mice bearing this construct with *CMV-Cre* mice resulted in the generation of the *Atad2* $\Delta 12$ allele, which can be verified through PCR amplification of the genomic DNA (shown on the ethidium bromide-stained gel). The expected amplified bands for each genotype allowing the detection of both wild-type and KO alleles of *Atad2* are indicated. E = Exon. (C) Extracts from wild-type and *Atad2* KO mice testes were probed with anti-ATAD2 and anti-H3 antibodies as indicated. Four different mice were used for this experiment. (D) Seminiferous tubules sections from wild-type and *Atad2* KO mice were stained with X-gal to visualize *lacZ* gene expression and β -galactosidase activity under the endogenous *Atad2* gene promoter. Testes from three different wild-type and *Atad2* KO mice were used to generate the represented sections. Scale bar: 200 μ m

on chromatin leading to its accumulation in ES cells, as well as in a variety of cancer cell lines (Wang et al. 2021 [↗](#)). Because of this very conserved action of ATAD2 in controlling the HIRA activity, we reasoned that the ATAD2-dependent HIRA activity should be particularly active during spermatogenic cell differentiation, especially, in post-meiotic cells, when ATAD2 expression is highly induced (Fig. 1D [↗](#)). To test if the absence of ATAD2 could lead to the accumulation of HIRA, as we previously reported in other systems, we quantified HIRA in spermatogenic cells from both wild-type and *Atad2* KO mice. These investigations revealed an accumulation of HIRA protein in *Atad2* KO spermatogenic cells (Fig. 2A [↗](#)), consistent with its increased association with chromatin (Wang et al. 2021 [↗](#)). We have also monitored HIRA accumulation as a function of spermatogenic cells stages. Because of the dramatic activation of *Atad2* after meiosis, in spermatids (Fig. 1D [↗](#)), the ATAD2-dependent accumulation of HIRA was expected to mostly occur in post-meiotic cells. Accordingly, we also compared HIRA accumulation in fractionated spermatogenic cell extracts. This experiment demonstrated that, in the absence of functional ATAD2, the accumulation of HIRA primarily occurs in round spermatids (Fig. 2B [↗](#)). This finding is consistent with the post-meiotic activation of *Atad2* and its role in regulating the turnover of chromatin-bound HIRA.

ATAD2 controls HIRA-dependent H3.3 localization and H3.3-dependent transcription

We speculated that an increased presence of HIRA on chromatin in the absence of ATAD2, should also be associated with increased amounts of deposited histones. Indeed, we have demonstrated previously in *S. pombe* that histone gene inactivation could rescue the growth arrest due to the absence of *Atad2/Abo1*. In parallel, we found that in ES cells, in the absence of ATAD2, an increased nucleosome assembly occurs, even on the so-called nucleosome-free region (NFR) (Wang et al. 2021 [↗](#)).

Interestingly, visualization of H3.3, the major target of HIRA by immunofluorescence in *Atad2* KO round spermatids, displayed a diffuse, intense staining (Fig. 3 [↗](#)), which is normally concentrated on the sex chromosomes. (van der Heijden et al. 2007 [↗](#); Fontaine et al. 2022 [↗](#)). Indeed, as expected and previously reported, in the wild-type round spermatids, H3.3 was mostly observed in a single domain, close to the unique chromocenter in these cells (Fig. 3 [↗](#), left panels). In the absence of ATAD2, in agreement with an increased residence time of HIRA on chromatin in general and a reduced H3.3 turnover, an increased level of H3.3 could be observed on the sex chromosome (adjacent to the chromocenter), and more interestingly, on all over the genome (Fig. 3 [↗](#), right panels).

General chromatin accessibility and transcriptional regulation by ATAD2

H3.3 is known to be dynamically incorporated on a group of active genes, especially around their transcriptional start sites (TSS) (de Dieuleveult et al. 2016 [↗](#)). We therefore predicted that the observed increased H3.3 residence time in the absence of ATAD2 would decrease TSS regions' accessibility specially those of highly active genes. To verify this prediction, we purified round spermatid fractions from wild-type and *Atad2* ko mice testes and measured the general chromatin accessibility following an ATAC-seq approach. Chromatin accessibility was then assessed around the TSS of genes, which were categorized into four groups based on the quartile of their mean expression from three independent transcriptomes of wild-type round spermatids (Fig. 4A [↗](#)), from the top 25% most highly expressed genes (group 1) to the 25% least expressed genes (group 4). As predicted, the absence of ATAD2 was associated with a significant decrease in TSS region accessibility of the most active genes (Fig. 4B [↗](#) and 4C [↗](#)), supporting a role for ATAD2 in maintaining chromatin dynamics at the most highly active chromatin regions.

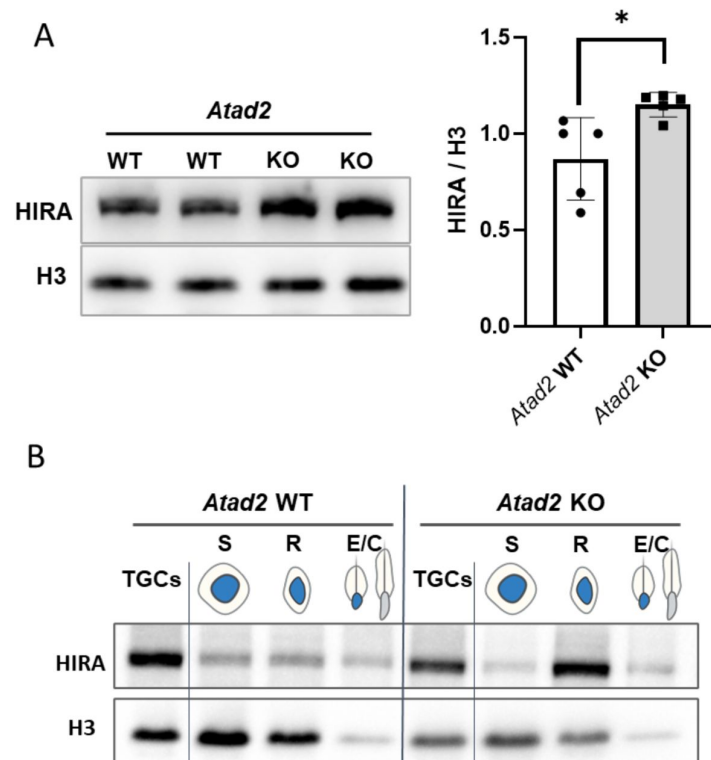


Figure 2

***Atad2* gene inactivation leads to the accumulation of HIRA.**

(A) Total testes' extracts from individual wild-type or *Atad2* KO mice (n =5) were used to detect HIRA and H3 by immunoblotting. The panel shows a representative immunoblotting after HIRA detection. The histogram shown in the lower left panel represents quantification of HIRA immunoblotting signal normalized to H3. Mean $\pm$ Standard Deviation is 0.86 ± 0.21 and 1.15 ± 0.06 for wild type and *Atad2* KO samples respectively and the p value is 0.02 for unpaired Student t-test. (B) Extracts from fractionated spermatogenic cells (pool of testes from 3 individuals per genotype) were used to detect HIRA and H3.

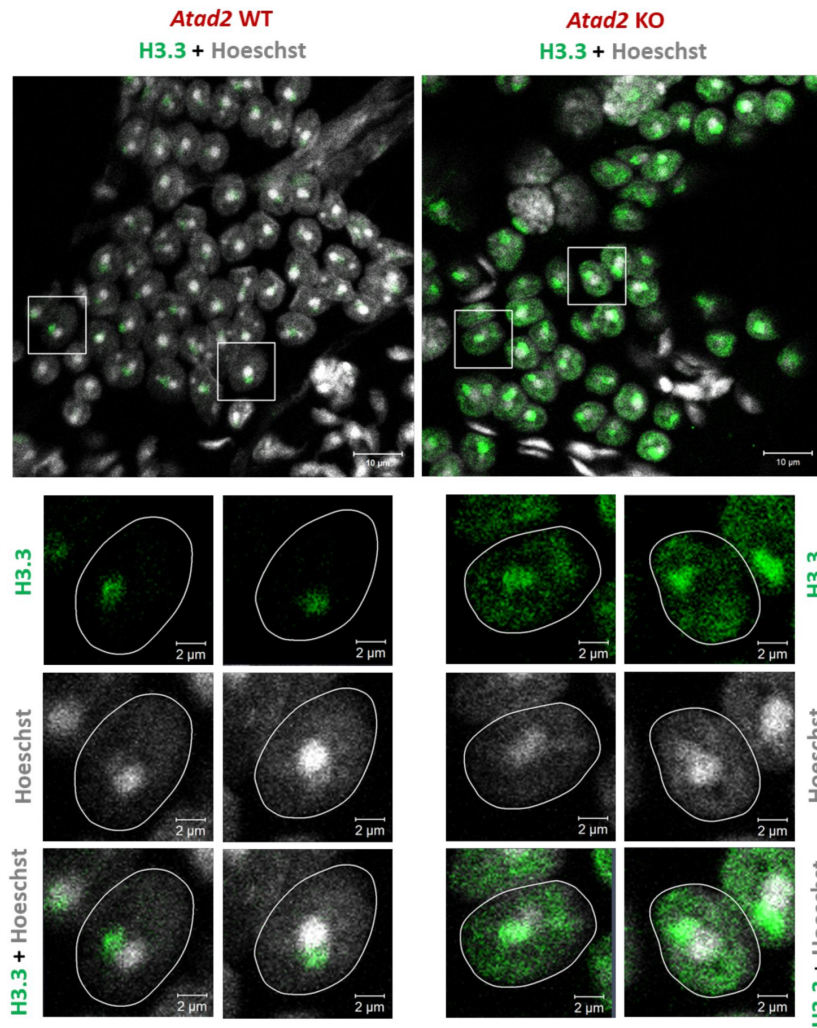


Figure 3

Enhanced H3.3 detection in *Atad2* KO post-meiotic cells A specific antibody against H3.3 was used to detect H3.3 in round spermatid cells from WT and *Atad2* KO mice.

Round spermatids are recognizable by their distinctive Hoechst - bright chromocenter (upper panels). The lower panels represent the magnification of selected cells shown at a lower magnification in the upper panels. The images show H3.3 labelling and Hoechst staining alone or the merged images of these staining as indicated.

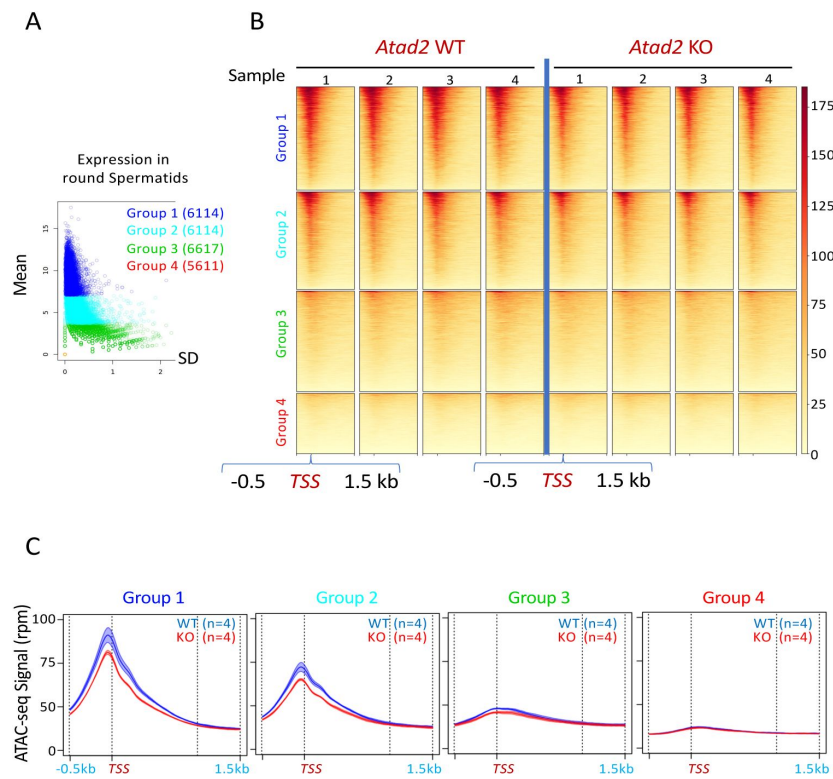


Fig. 4

ATAD2 controls active gene TSS accessibility.

A) Gene selection according to expression in round spermatids. The plot displays the mean expression levels on the y-axis and the corresponding standard deviations on the x-axis, calculated from three independent biological replicates of isolated round spermatids. The genes were divided into 4 groups (grp1 in blue, grp2 in cyan, grp3 in green and grp4 in orange) according to the quartile of their mean expression. **B) Heatmap on ATAC-seq signal.** The heatmap shows the ATAC-seq signal of each group (grp1 in blue, grp2 in cyan, grp3 in green and grp4 in orange) in rows, for 4 ATAD2 WT and 4 ATAD2 KO samples in column. **C) Profiles according to selected genes.** The panels show for each group the mean profile (± 2 s.e.m) (solid line) of four ATAD2 WT replicates (in blue) and 4 ATAD2 KO (in red) (transparency) over the four replicates.

To visualize the genes more sensitive to ATAD2-mediated chromatin dynamics affecting transcription, we monitored whole genome transcriptional activity during the first spermatogenic wave at day 20 post-partum (PP), when meiosis is complete, and the first post-meiotic cells appear on 22-, 24- and 26-days PP.

In order to definitively confirm our working hypothesis on the ATAD2-HIRA-H3.3 functional interaction, we took advantage of published H3.3-dependent transcriptomic data (Fontaine et al. 2022 [\[4\]](#)). Indeed, the authors of this publication identified a series of genes that are mis-regulated in cells with inactivated *H3.3b* gene. Because of the post-meiotic major activation of *Atad2*, we specifically focussed on the gene lists that show a H3.3-dependent deregulated expression in post-meiotic round spermatids.

In these transcriptomes, we specifically followed the expression of the published lists of genes that are downregulated or upregulated in the absence of H3.3 (Fontaine et al. 2022 [\[4\]](#)). Very interestingly, we observed that the majority of genes that were reported to be down-regulated in the absence of H3.3 (H3.3-activated genes), are upregulated in the absence of ATAD2, particularly at day 26 PP, when a significant number of post-meiotic cells are present in the testes (**Fig. 5A** [\[4\]](#), left and middle panels). Gene Set Enrichment Analysis (GSEA) applied to genes that are differentially expressed between wild-type and *Atad2* KO cells at day 20 PP, confirmed that the “H3.3-activated” gene set (corresponding to the published list of H3.3-activated genes) are clearly enriched in *Atad2* KO cells (**Fig. 5A** [\[4\]](#), right panel).

This regulation can also be observed when we considered the published list of genes that were up-regulated in the absence of H3.3 (H3.3-repressed genes). In this case, the absence of ATAD2 led to the enhancement of their down-regulation. In other words, the absence of ATAD2 led to an enhancement of H3.3-dependent gene repression (**Fig. 5B** [\[4\]](#)).

Fontaine and colleagues also reported that sex chromosome-linked genes that normally undergo Meiotic Sex Chromosome Inactivation (MSCI), were up-regulated in the absence of H3.3, indicating that H3.3 is required for their repression. We also investigated these genes and showed that they present an enhancement of repression in the absence of ATAD2 (**Fig. 5C** [\[4\]](#)).

These analyses clearly indicate that the absence of ATAD2 enhances H3.3 functions in agreement with increased residence time of H3.3 in nucleosomes. Genes that require H3.3 to be repressed or activated present an enhanced repression or activation, respectively, in the absence of ATAD2. Overall, these investigations remarkably confirmed a role for ATAD2 in regulating H3.3 function.

ATAD2 is required for controlled histone eviction and PRM assembly

ATAD2's involvement in HIRA-chromatin interaction dynamics and its role in H3.3 assembly in post-meiotic cells, along with its similar function in the dynamics of other histone chaperones like FACT (Wang et al. 2021 [\[4\]](#)), led us to investigate whether ATAD2 participates in controlling the histone-to-protamine replacement occurring later in these cells. The histone eviction process involves a genome-wide histone H4 hyperacetylation (Shiota et al. 2018 [\[4\]](#)) and a TP-mediated PRMs assembly, which directs histone eviction and the final packaging of the male genome (Barral et al. 2017a [\[4\]](#)). To test the involvement of ATAD2 in spermatogenic cell chromatin reorganization, we sought to co-detect testis-specific H2B variant, TH2B (Montellier et al. 2013 [\[4\]](#)), along with transition protein 1 (TP1) and protamine 1 (PRM1). TH2B is first expressed at the commitment of spermatogenic cells into meiotic divisions and progressively becomes the major H2B species of spermatogenic cell chromatin in the subsequent stages (Montellier et al. 2013 [\[4\]](#)). As expected, the accumulation of TP1 and PRM1 in wild-type cells is associated with the eviction of the majority of histones, reflected by the mutually exclusive labelling of TH2B and TP1/PRM1 in distinct cells (**Fig. 6A** [\[4\]](#) and **6B** [\[4\]](#), *Atad2* wild-type mice). In contrast, in *Atad2* KO cells, TH2B and TP1/PRM1 were

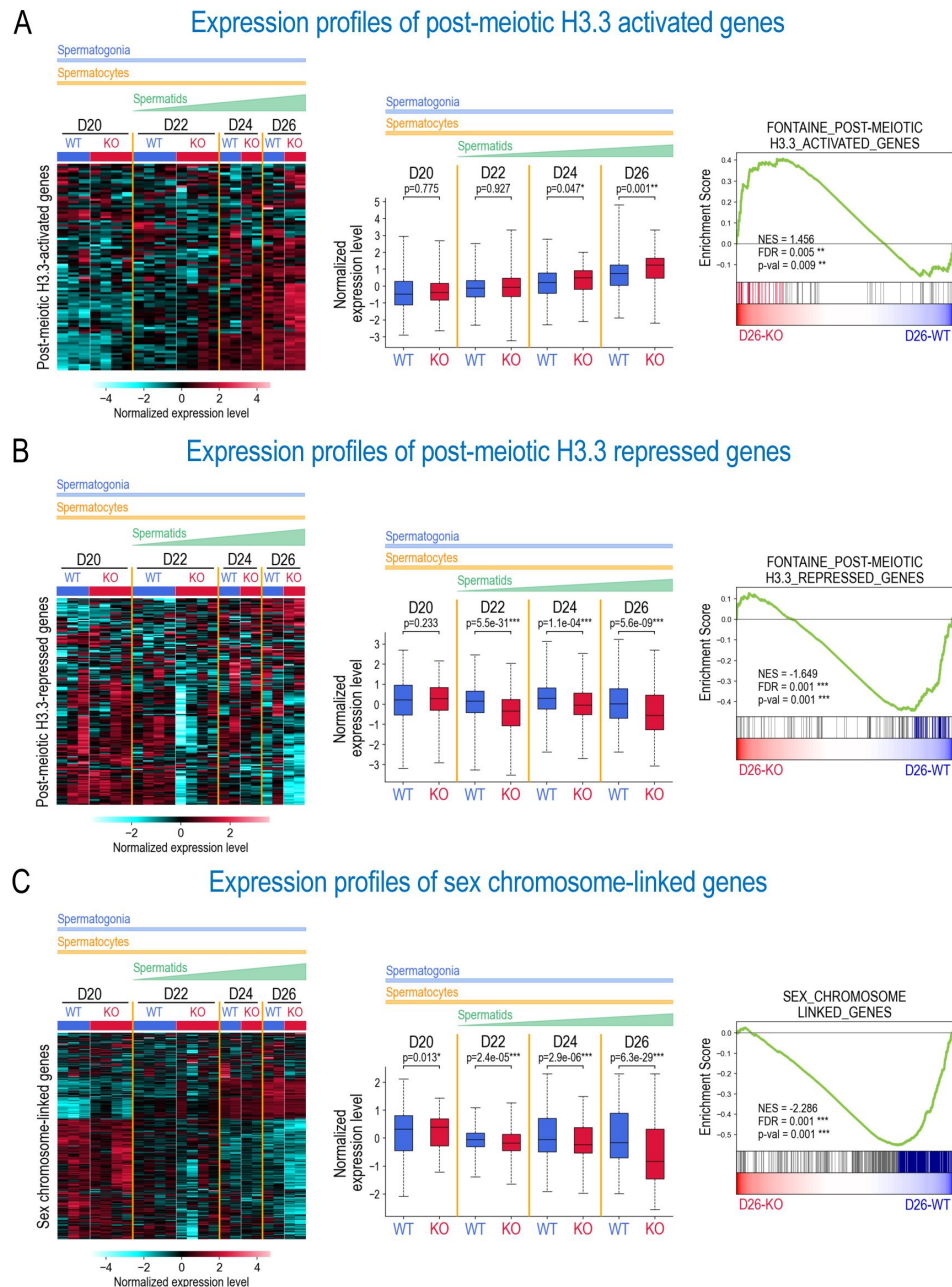


Figure 5

ATAD2 enhances H3.3 gene regulatory functions.

(A) Expression profiles of post-meiotic H3.3-activated genes. The heatmap (left panel) displays the normalized expression levels of genes identified by Fontaine and colleagues as upregulated in the absence of histone H3.3, (Fontaine et al. 2022 [\[1\]](#)) for ATAD2 WT and ATAD2 KO samples at days 20, 22, 24, and 26 PP. The colour scale represents the z-score of log-transformed DESeq2-normalized counts. The box plots (middle panel) show the pooled normalized expression levels, aggregated across replicates and genes, for each condition (WT and KO) and time point (D20 to D26). Statistical significance between WT and KO conditions was determined using a two-sided t-test, with p-values indicated as follows: * for p-value<0.05, ** for p-value<0.01, *** for p-value<0.001. The GSEA results (right panel) highlight a significant enrichment of the “post-meiotic H3.3-activated genes” gene set in KO samples relative to WT samples at D26. The coloured vertical bars represent the core genes that account for the gene set’s enrichment signal. **(B)** As in (A) for the “post-meiotic H3.3-repressed genes” gene set. **(C)** As in (A) for the “sex chromosome-linked genes” gene set.

co-detected in a significant number of cells (**Fig. 6A** and **6B**), suggesting that the process of histone eviction is disturbed. However, in *Atad2* KO cells, the initial delay in histone removal is transient, as spermatids with high levels of transition protein TP1 and PRM1 eventually appear (**Fig. 6C**).

These observations strongly suggest that, in the absence of ATAD2, the process of histone eviction is also slowed down or delayed, allowing for the simultaneous detection of histones and histone-replacing proteins in the same cells.

Given that NUT-CBP/p300-dependent H4 hyperacetylation, along with the expression of H2A.L.2, TPs, and PRMs, is essential for histone removal, we also investigated whether the observed delay in histone removal is due to defective histone acetylation, particularly at H4K5, as well as disrupted cell type-specific expression and localization of H2A.L.2, TPs, and PRMs. Immunohistochemical detection of H4K5ac, TP2, PRM1, and PRM2 revealed a similar pattern and cell type-specific expression of the investigated proteins when comparing *Atad2* KO and wild-type post-meiotic cells. (**Fig. S1**).

These data support the idea that the perturbed modified dynamics of histone/TP/PRM chaperones does not grossly affect the other molecular events involved in the histone-to-PRM replacement.

Perturbed pre-PRM2 processing in the absence of ATAD2

We have shown previously that the H2A.L.2-dependent process of incorporation of TPs into nucleosomes is required for the proper processing of pre-PRM2. The abnormal overlapping staining of histones, TP and protamines observed by immunofluorescence in *Atad2* KO cells, suggested that TPs may not effectively mediate the pre-PRM processing. To test this hypothesis, we took advantage of our specific pre-PRM2 antibody (Rezaei-Gazik et al. 2022) to investigate pre-PRM2 maturation and assembly. **Figure 7A** shows that although pre-PRM2 is present in both wild-type and *Atad2* KO elongating spermatids, in the absence of ATAD2, pre-PRM2 tends to form dense aggregates as well as fragmented genomic domains. This phenomenon is also associated with an increased accumulation of pre-PRM2 protein level in testis extracts of *Atad2* KO spermatogenic cells compared to wild-type cells as revealed by ELISA (**Fig. 7B**).

These data provide strong evidence that ATAD2 is required to streamline the activity of histone chaperones as well as of factors involved in histone exchange and TP/PRM assembly.

Defective mature spermatozoa genome organization and male fertility *in vitro*

We show that although the *Atad2* KO post-meiotic cells' defective chromatin organization does not affect the testis histology (**Fig. S1** and data not shown), it impairs testis weight and sperm count, which are significantly reduced in *Atad2* KO mice (**Fig. S2**).

As expected, the impact of ATAD2 on HIRA function and TP/PRM assembly also has an impact on proper sperm genome compaction. Indeed, mature sperm genome decompaction test revealed a significant difference in the ability of the genome to resist genome decompaction conditions between wild-type *Atad2* and *Atad2* KO spermatozoa (**Fig. 8A**). Given that spermatozoa with defective genome compaction are known to have poor success in *in vitro* fertilization (Bashiri et al. 2021), we expected that spermatozoa from *Atad2* KO mice would perform worse in *in vitro* fertilization assays compared to wild-type spermatozoa. **Figure 8B** shows that this is, in fact, the case. However, surprisingly, fertility tests carried out on sexually mature adult males revealed that *Atad2* KO male mice are fertile, with only a statistically non-significant effect observed on their ability to foster pups (**Fig. 8C**), indicating that *in vivo* fertilization could buffer a relatively large series of sperm genome compaction defaults.

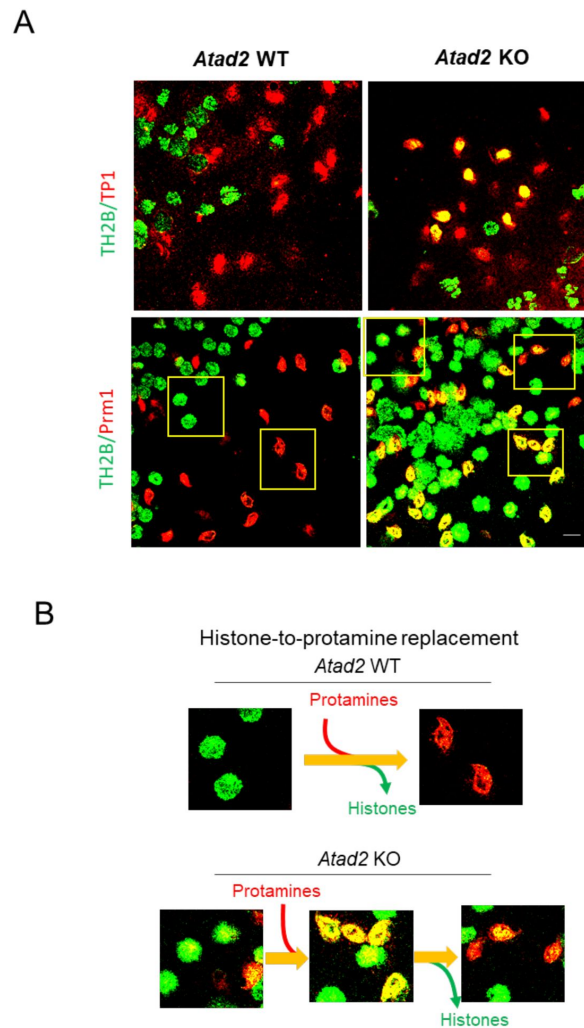


Figure 6

***Atad2* is required for efficient histone-to-protamine replacement.**

(A) Spermatogenic cell preparations from wild-type and *Atad2* KO mice were stained with antibodies against TH2B (in green) and transition protein 1 (TP1) or protamine 1 (PRM1) (in red). A yellow signal indicates the co-detection of TH2B along with TP1 or PRM1. The indicated fields are shown at higher magnification in panel “B”. Scale bar: 10 μ m. (B) Fields shown in panel “A” are ordered to illustrate the transition from the histone associated genome to its packaging by protamines, considering both wild-type and *Atad2* KO spermatogenic cells.

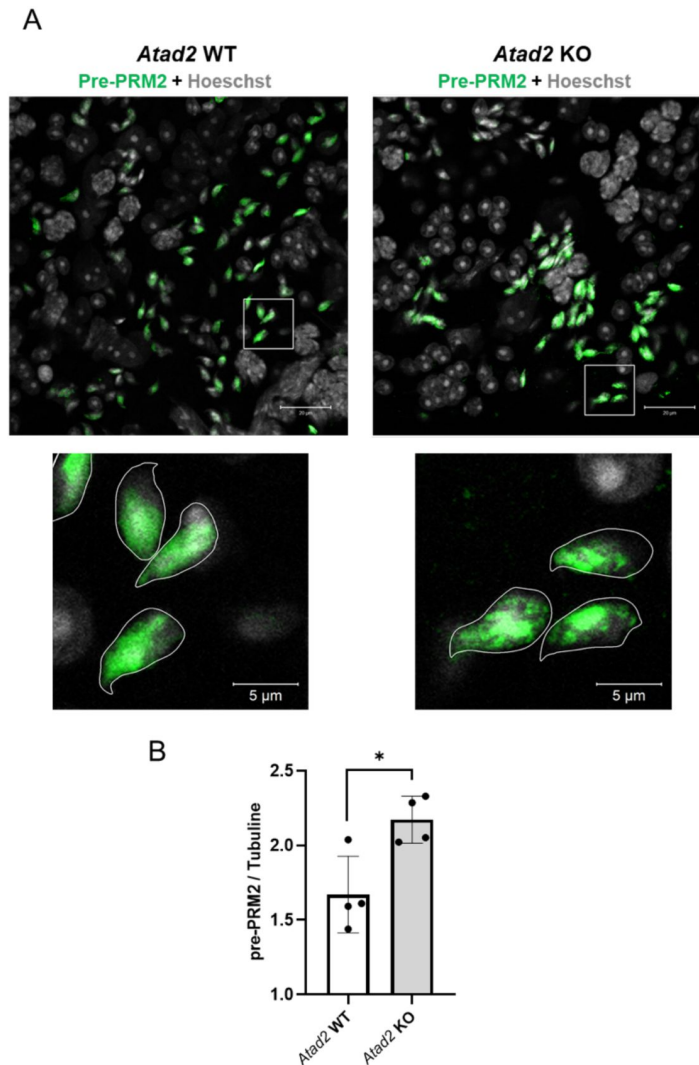


Figure 7

***Atad2* is required for proper pre-protamine 2 processing and protamine assembly**

(A) A specific antibody raised against the processed part of pre-PRM2 (Rezaei-Gazik et al. 2022 [\[1\]](#)) was used to detect pre-PRM2 expression and localization in tubular elongating spermatid cells. The lower panels represent the magnification of selected cells. (B) The amount of pre-PRM2 was detected using soluble extracts from testis of wild-type and *Atad2* KO mice using ELISA. Histograms show the normalized values (pre-PRM2/tubulin) of 4 measurements of sedimentation samples of Elongating/Condensing fractions. Mean ± Standard Deviation is 1.6 ± 0.2 and 2.1 ± 0.1 for wild type and *Atad2* KO samples respectively and the p value equals 0.0158 when applying unpaired Student t-test.

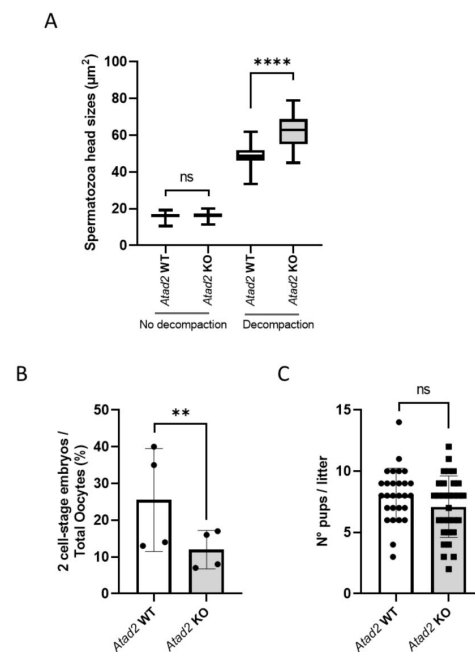


Figure 8

ATAD2s' function in sperm genome compaction and fertility parameters

(A) Epididymal sperm cells were isolated by the swim-up method from wild-type or *Atad2* KO mice and underwent a decompaction test and the sizes of the sperm heads were measured and represented as box plots ($n = 75$ sperm heads / box plot). The value ranges (mean) were 10.4 ~ 19.3 (16.0) for wild type sperms without decompaction; 11.4 ~ 20.1 (16.4) for *Atad2* KO sperms without decompaction; 33.4 ~ 61.8 (48.5) for wild type sperms after decompaction; 45.0 ~ 78.9 (62.6) *Atad2* KO sperms after decompaction. Tukey's HSD post two-way ANOVA test was used. **** $p < 0.001$. (B) Pools of oocytes from C57BL6 females were obtained and used for *in vitro* fertilization (IVF) with spermatozoa from WT or *Atad2* KO males (the experiment was repeated 4 times independently). The fertilization success rate is indicated as a percentage of oocytes giving rise to stage 2 embryos. Mean $\pm$ Standard Deviation is 25.5 ± 14 and 12 ± 5.2 for wild type and *Atad2* KO experiments respectively and the $p = 0.002$ for ratio paired Student t-test. (C) Wild-type ($n = 5$) and *Atad2* KO ($n = 5$) male mice were crossed with C57BL6 female mice and the number of pups was counted and reported as box plots ($n = 28$ and 30 , respectively). The average numbers of pups are respectively 8.0 ± 2.1 and 7.1 ± 2.5 and the p -value of Student t-test = 0.1.

Discussion

Taking advantage of the very conserved structural features of ATAD2 and relying on the power of yeast genetics, we previously discovered that ATAD2's function is mostly to control major histone chaperone functions. Using a screen of the Abo1/ATAD2-dependent arrest of *S. pombe* cell growth, we found that impressively, 11 of the 14-growth arrest suppressor strains that we isolated, impacted histone chaperones or histones themselves, with 8 distinct inactivating mutations in the histone chaperone HIRA complex. Furthermore, investigation of ATAD2-depleted mouse ES cells showed that HIRA accumulates and remains bound to nucleosomes, leading to nucleosome overloading around active gene TSSs and on the normally nucleosome-free regions (NFRs) of these genes.

Here, we showed that *Atad2* is highly expressed in post-meiotic spermatogenic cells and that in its absence, as in ES cells and in different cancer cells, HIRA accumulates. Our new data suggest that in the absence of ATAD2 activity, HIRA residency time on chromatin increases leading to decreased dynamics and overloading of its substrate, the histone variant H3.3. This replication-independent histone has been first shown to assemble on the transcriptionally inactive meiotic cell sex chromosomes by Peter de Boer's laboratory ([van der Heijden et al. 2007](#)). Subsequent studies showed that H3.3 remained enriched on the sex chromosomes also in early post-meiotic cells, usually localized close to the unique large chromocenter in these cells ([Fontaine et al. 2022](#)).

In agreement with the proposed role for ATAD2 in the control of HIRA-dependent dynamics of H3.3, we observed a more intense labelling of H3.3 on the sex bodies in round spermatids and impressively, an increase in the genomic background detection of H3.3 in these cells.

The role of ATAD2 in the HIRA/H3.3 functions is further clearly supported by comparing the reported H3.3-dependent transcriptome with the ATAD2-dependent transcriptome identified here.

Indeed, in the absence of ATAD2, we observed an enhancement of H3.3-dependent gene regulatory events (activation or repression), in agreement with an increased H3.3 presence and hence function, in the chromatin of *Atad2* KO post-meiotic cells.

These investigations on ATAD2's role in post-meiotic spermatogenic cells built upon our previous findings in *S. pombe* and ES cells, uncover an additional level of regulation of histone dynamics, that relies on the dynamics of histone chaperone - chromatin interaction. Previously, A. Groth's laboratory discovered that histone chaperone DNAJC9 recruits the ATPase HSP70 to provide the necessary energy to release the chaperone from chromatin ([Hammond et al. 2021](#)). Considering our data on the role of the ATPase ATAD2 with respect to HIRA and that of HSP70 with respect to DNAJC9, it is tempting to propose that ATP energy is required for the proper function of histone chaperones in controlling histone turnover.

Interestingly, very recently, Song laboratory demonstrated that the *S. pombe* Abo1/ATAD2, is required for ATP-dependent dissociation of the histone chaperone FACT from chromatin, particularly at TSSs ([Jang et al. 2024](#)). In line with this conclusion, ChIP-seq mapping of FACT in *Atad2*-depleted ES cells revealed an accumulation of nucleosome-bound FACT at the TSSs of active genes ([Wang et al. 2021](#)), supporting the idea that ATAD2 not only regulates the dynamics of histone-bound HIRA but also that of other histone chaperones such as FACT

In addition to histones, many chromatin-bound regulatory factors also need to dynamically interact with chromatin and hence ATAD2 could also be involved in mediating the dynamic interactions of such chromatin-bound factors. In agreement with this hypothesis, using proteomics of ATAD2-bound nucleosomes in ES cells, we previously co-purified various chromatin bound

factors (Morozumi et al. 2016 [\[3\]](#)). Other AAA ATPase factors such as p97/VCP, have also demonstrated actions in the dynamics of chromatin-bound factors. Indeed, the AAA ATPase factor, p97/VCP, has been shown to remove H3 variant CENP-A (van den Berg et al. 2023 [\[3\]](#)), the chromatin trapped PARP1 (Krastev et al. 2022 [\[3\]](#)), trapped Ku70/80 (van den Boom et al. 2016 [\[3\]](#)), DNA damage sensors, DDB2 and XPC (Puumalainen et al. 2014 [\[3\]](#)), RNA pol II (Lafon et al. 2015 [\[3\]](#)), and other chromatin interacting factors. Similarly, ATAD2 AAA ATPase very likely acts on other chromatin-bound factors than histone chaperones. However, in contrast to p97/VCP, and due to its bromodomain, ATAD2 would preferentially act on H4K5 acetylated chromatin regions and would specifically ensure the dynamics of H4 acetylated chromatin-bound factors. Based on the data presented here, we can also include, a role for ATAD2 to streamline steps involving the H2A.L2-dependent recruitment of TPs and protamine assembly and the final histone displacement in spermatids. The chaperones responsible for TP/PRM assembly are not yet identified, but it is possible that ATAD2 plays a role in dissociating these chaperones from their substrate proteins after the assembly process. That is why in our *Atad2* KO spermatids, we observed an overlap between TP/PRM and histones. This could be due to the delayed removal of chaperones bound to TPs and PRMs and the inability of pre-PRM2 to be processed and hence of PRMs to efficiently replace histones. Since histone replacement still occurs in the absence of ATAD2, it suggests that other AAA ATPases, such as p97/VCP, may compensate for its function. However, under these conditions, although most of the histone replacement takes place, our functional tests such as *in vitro* chromatin decompaction and *in vitro* fertilization, strongly suggest that mature spermatozoa genome organization is not fully complete and the spermatozoa nuclei remain fragile and sensitive to external assaults.

In summary, the functional analysis of ATAD2 during its physiological expression in the post-meiotic phases of spermatogenesis supports our previous conclusions regarding its role in ES cells and its conserved activity in *S. pombe*. Together, these studies reveal a novel layer of regulation in histone turnover and chromatin dynamics, based on the control of histone-bound histone chaperone release.

Materials and methods

Animal care and breeding

Mice were housed at the Grenoble High Technology Animal Facility (PHTA). Mice were euthanized following a procedure approved by the official ethic committee of the University Grenoble Alpes (COMETH, C2EA-12). The investigators directly involved in care and breeding of mice had an official animal-handling authorization obtained after 2 weeks of intensive training and a final formal evaluation.

Generation of *Atad2*-KO mice

Atad2 KO mice were obtained from the International Knockout Mouse Consortium, the Wellcome Trust Sanger Institute, UK. The 'knockout-first' allele (tm1a) contains an IRES: *lacZ* trapping cassette and a floxed human β -actin promoter-driven *neo* cassette inserted into the intron 12 of the *Atad2* gene (Skarnes et al. 2011 [\[3\]](#)). We created *Atad2* ^{Δ Neo-Exon12 Neo-Exon 12} mice by crossing mice bearing the *tm1a* allele with mice expressing Cre recombinase under the control of a ubiquitous human promoter CMV. This cross results in an *Atad2*^{*LacZ* Δ Neo-Exon12} knockout mouse with the first 11 exons of *Atad2* being fused to β -gal, leading to a non-functional *Atad2* protein depleted of its bromodomain and AAA ATPase domain.

For *Atad2* KO mice genotyping, multiplex PCR was used to detect *LacZ* cassette producing a 393-bp WT band and a 293-bp mutant band using primers *Atad2*_84231_F, TATCCAACAAGCCTGAGCCC, *Atad2*_84231_R, CAACTGGAGCTGGGTCTTCC and Cas-R, TCGTGGTATCGTTATGCGCC.

Fertility test

Fertility tests were carried out with sexually mature 3-month-old males. Five WT males and five *Atad2* KO males were crossed with two C57BL6 females each, during 3 months, and the litter sizes were monitored during this period.

In vitro Fertilization

Eggs were obtained from 6- to 7-week-old superovulated C57/BL6J females, stimulated by a first injection of 7.5 UI of PMSG (Pregnant Mare Serum Gonadotropin), followed 48 hours later by a second injection of 7.5 UI of HCG (Human Chorionic Gonadotropin). Fourteen hours later, females were euthanized by cervical dislocation and cumulus-oocytes complexes (COCs) released from the ampulla were collected in 500 μ L M2 medium. Spermatozoa from wild type and *Atad2* KO males were harvested by dilaceration of the cauda epididymitis and allowed to swim in 1 mL M2 medium for 10 min at 37°C. Two hundred microliters of sperm were transferred into 800 μ L M16 /2% BSA and capacitated for 80 min at 37°C /5% CO₂. Finally, 300,000 sperm were simultaneously added to the COCs and incubated in M16 medium at 37°C/5% CO₂. After 4 hours of incubation, unbound spermatozoa were removed by 3 successive washes with 500 μ L of M16. Twenty-four hours after fertilization, unfertilized eggs and two-cell embryos (as an indication of successful fertilization), were scored.

Protein extraction and Immunoblotting

Proteins were extracted in Urea 8M, sonicated for 150J with a probe sonicator, mixed with 4x Laemli loading buffer and heated for 5 minutes at 95 °C. Protein samples were loaded on 4-12% Bis-Tris gels and migrated at 150 V for 1h30. The transfer step was realized using a nitrocellulose membrane with the Bio-Rad Transfer Turbo machine. Nitrocellulose membrane was blocked in 5% milk PBS-Tween 0.1% solution, washed 3 times for 10 minutes with PBS-Tween 0.1% solution and incubated overnight at 4°C with specific antibodies listed in [Table S1](#). Membranes were washed 3 times for 10 minutes with PBS-Tween 0.1% solution and were incubated with HRP secondary antibodies at 1/10000 dilution for 1h at RT under agitation. Revelation was performed using ECL substrates and visualization was performed with the Vilber Fusion FX Chemiluminescence imaging system (Vilber).

ELISA

Urea protein extracts were diluted in PBS and used to coat 96-well round bottom plates by incubating at 4°C overnight. Coating solution was removed and wells were washed 3 times with washing buffer (BSA 1%, PBS-T 0,1%). Wells were blocked with blocking buffer (5% BSA in PBS-T 0,1%) for 2 hours at room temperature under agitation. Blocking buffer was removed and wells were washed 2 times with washing buffer. Primary antibodies for pre-PRM2 (1:1000) and tubulin (1:5000) were diluted into blocking buffer, pipetted into their corresponding wells and incubated overnight at 4°C under mild agitation. Wells were washed 4 times with washing buffer and subsequent incubation with secondary HRP antibodies, diluted 1:5000 in blocking buffer, was performed for 1 hour at room temperature under mild agitation. Wells were washed 4 times with washing buffer, and TMB ELISA Substrate was added to wells and incubated until the color developed, at which time Stop Solution was added to the wells to stop the reaction. The signal was quantified using a plate reader at 450nm.

X-Gal staining

Male mice were euthanized by cervical dislocation. Testes were dissected out, drilled in 3 points and incubated overnight at 4°C under agitation in fixative solution (0.5% glutaraldehyde, 2 mM MgCl₂ and 5 mM EGTA in PBS). After incubation until equilibration in an equilibration buffer (30% sucrose, 2 mM MgCl₂ and 5mM EGTA in PBS), testes were frozen individually in OCT on dry ice.

Frozen blocks were sectioned at 10 μm slices on SuperfrostTM slides and let dry overnight. Slides were fixed with fixation buffer (0.2% glutaraldehyde, 100 mM MgCl_2 and 5 mM EGTA in PBS) for 10 min, washed with washing buffer (PBS, 0.02% NP-40, 0.01% Sodium deoxycholate and 2 mM MgCl_2) for 5 min and incubated three times in 50% Ethanol for 5 min each. Then the slides were stained with staining solution (2 mM MgCl_2 , 0.02% NP-40, 0.01% Na Doc, 5 mM potassium ferricyanide, 10 mM potassium ferrocyanide and 0.5 mg/ml X-gal) in a wet chamber for several hours at 37°C. The slides were then washed with washing buffer for 5 min and with distilled water, and a counterstaining was performed with hematoxylin. Finally, the slides were washed with running tap water for several minutes and mounted with Dako fluorescent mounting medium.

Immunofluorescence staining of germ cells

Staged seminiferous tubules were prepared as detailed in [Gaucher et al. 2012](#). In order to permeabilize the cells, the slides were then placed in 0.5% saponin, 0.2% Triton X-100, and 1 $\times$ PBS at RT for 15 min and washed for 5 min with PBS. The preparations were then incubated in 5% milk, 0.2% PBS-Tween blocking buffer, at RT for 30 min under agitation. The primary antibodies were diluted in 1% dry milk, 0.2% PBS-Tween buffer. The slides were incubated overnight with this diluted solution of the primary antibodies in a humidified chamber at 4°C and then washed three times for 5 min each in the antibodies' dilution buffer. The secondary antibodies were diluted at 1:500 in the same buffer and incubated in a humidified chamber for 30 min at 37°C, and then washed as for the primary antibodies. The DNA was counterstained by Hoechst, and the slides were mounted using Dako fluorescent mounting medium.

Sperm decompaction test

Spermatozoa from wild type and *Atad2* KO males were harvested by dilaceration of the cauda epididymis and allowed to swim in 1 mL M2 medium for 10 min at 37°C. The sperm-containing suspension was transferred into a 1.5 ml tube and centrifuged at 3000 rpm for 8 min in 4°C. The sperm pellets were resuspended in PBS and dropped onto glass slides to spread and dry at RT. The decompaction mix (50 mM DTT, 400 IU/ml heparin, 0.2% Triton in PBS) was added to each dried sperm drop for 2 min and the decompaction was stopped by placing the slides in 4% PFA for 15 min. After washing with PBS for 5 min, the slides were stained with DAPI and mounted using Dako fluorescent mounting medium. Fluorescent images were captured by a Zeiss inverted microscope under a 63X numeric aperture oil-immersion lens (Carl Zeiss). The size of spermatozoa heads was calculated with ImageJ software.

Spermatogenic cell fraction purification

Spermatogenic cell fraction purification was performed as previously described ([Shiota et al. 2018](#)).

ATAC-seq and data analysis

Library preparation

Frozen nuclei from four biologically independent purified spermatid fractions were centrifuged at 3000 g for 5 minutes and then washed in a 1% BSA-PBS solution. Libraries were then prepared from 100,000 nuclei using the ATAC-seq Kit (Catalog No. 53150) from Active Motif, following the manufacturer's protocol (version B9). After quality control (QC) with Qubit and Fragment Analyzer, libraries were subsequently purified using AMPure XP (Beckman Coulter, Catalog No. A63881) beads with a double size selection of 100-800 bp fragments. After a second round of library QC (Qubit and Fragment Analyzer) we performed a paired-end sequencing on a NextSeq 2000 device using a 200-cycle P2 flow cell.

Trimming

The raw fastq files were processed by 5' end trimming, keeping 30bp-length fragments, using fastx_trimmer [http://hannonlab.cshl.edu/fastx_toolkit/]. Accessed 28 Feb. 2022.], with options -l 30 -Q33.

Alignment

The trimmed fastq files were aligned on the UCSC Mus_musculus mm10 genome using the Bowtie2 aligner (Langmead and Salzberg 2012), with options -end-to-end, -no-mixed, -no-discordant.

Normalization

The big wig files (.bw) containing normalized integrated aligned read count signals were obtained from by normalizing and smoothing bam files using bamCoverage (from deepTools suite (Ramirez et al. 2014)) with options: -binSize 4 -minMappingQuality 30 -normalizeUsing RPM.

Heatmap

The normalized ChIP signal were converted into a 10bp bin matrix of the signal 500b upstream and 1500b downstream protein-coding genes TSS, using computeMatrix (from deepTools suite (Ramirez et al. 2014)), with options reference-point -R mm10_pc.bed -referencePoint TSS -binSize 10 -beforeRegionStartLength 500 -afterRegionStartLength 1500 Heatmaps were generated using plotHeatmap (deepTools suite (Ramirez et al. 2014)).

RNA-seq data processing

Wild-type (WT) and *Atad2* knockout (KO) homozygous male pups were sacrificed by cervical dislocation on days 20-, 22-, 24-, and 26-postpartum (PP), after which their testes were harvested. RNA was extracted using TRIzol™ reagent, followed by DNA digestion with DNase I. The resulting RNA samples were used for RNA sequencing. For both WT and KO mice, the biological replicates included 3 samples for D20, 4 samples for D22, 2 samples for D24 and 2 samples for D26.

The raw sequencing data (FASTQ files) were aligned to the UCSC mm10 mouse genome using the STAR software (version 2.7.11b) (Dobin et al. 2013) to generate BAM files. Raw read counts were extracted from the BAM files using HTSeq (version 2.0.5) (Anders et al. 2015) with the following parameters: -t exon, -f bam, -r pos, --stranded=reverse, -m intersection-strict, and --nonunique none.

Normalization of read counts was performed using the DESeq2 R package (version 1.22.2) (Anders and Huber 2010; Love et al. 2014). Counts were transformed into DESeq2-normalized values and subsequently log-transformed using the formula $\log_2(1 + \text{DESeq2-normalized counts})$. Differential expression analysis was conducted with the R package “SARTools” (Varet et al. 2016).

Purified RNA from fractionated round spermatids were also sequenced to establish gene transcriptional group quartile used shown in Fig. 4. RNA purification and sequencing were performed as described above.

Quantification and statistical analysis

Statistical analysis was performed with R and GraphPad Prism 8. The results of fertility tests were analyzed by Student t-test. Pairwise Student t-test was used to analyze testis weights and spermatozoa counts of same-aged mice and ratio pairwise Student t-test was used for the IVF

efficiency results. The results of sperm decompaction test were analyzed by two-way ANOVA and Tukey's HSD post two-way ANOVA test was used to determine the significance between groups. *P-values* < 0.05 were considered statistically significant: **p* < 0.05, ***p* < 0.01, ****p* < 0.001.

Gene Set Enrichment Analysis (GSEA)

Differential expression analysis and the corresponding GSEA analysis (Mootha et al. 2003 [↗](#); Subramanian et al. 2005 [↗](#)) were performed to compare *Atad2* KO versus *Atad2* wild-type samples at days 20, 22, 24, and 26 PP. For the GSEA analysis, we used the Python package “gseapy” (Fang et al. 2023 [↗](#)) available at <https://www.gsea-msigdb.org/gsea> [↗](#).

The GSEA was carried out on three custom gene sets. The first gene set, referred to as “sex chromosome-linked genes”, corresponds to genes located on the X and Y chromosomes. The remaining two gene sets include genes identified as upregulated or downregulated in the absence of histone H3.3 by Fontaine and colleagues (Fontaine et al. 2022 [↗](#)). These gene sets were designated as “post-meiotic H3.3-activated genes” and “post-meiotic H3.3-repressed genes”, respectively.

Data availability

The transcriptomic data generated in this study, corresponding to testes collected at different days post-partum, have been deposited on GEO under the accession No. GSE277943.

To review GEO accession GSE277943 go to: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE277943> [↗](#) Enter token wrszesqgjfkxxmx into the box.

Transcriptomic data from fractionated round spermatids (three biological replicates, **Fig. 4A** [↗](#)) were deposited on GEO accession GSE284749. To review GEO accession GSE277943 go to: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE284749> [↗](#). Enter token uzmvcaypgnydrgrt into the box ATAC-seq data generated in this study were deposited on GEO under the accession N° GSE283879. To review GEO accession GSE283879, go to: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE283879> [↗](#) Enter token ozazwsmoxjofzkb into the box Immunoblot raw data used in preparing the figures are presented as Supplementary file 1

Supplementary materials

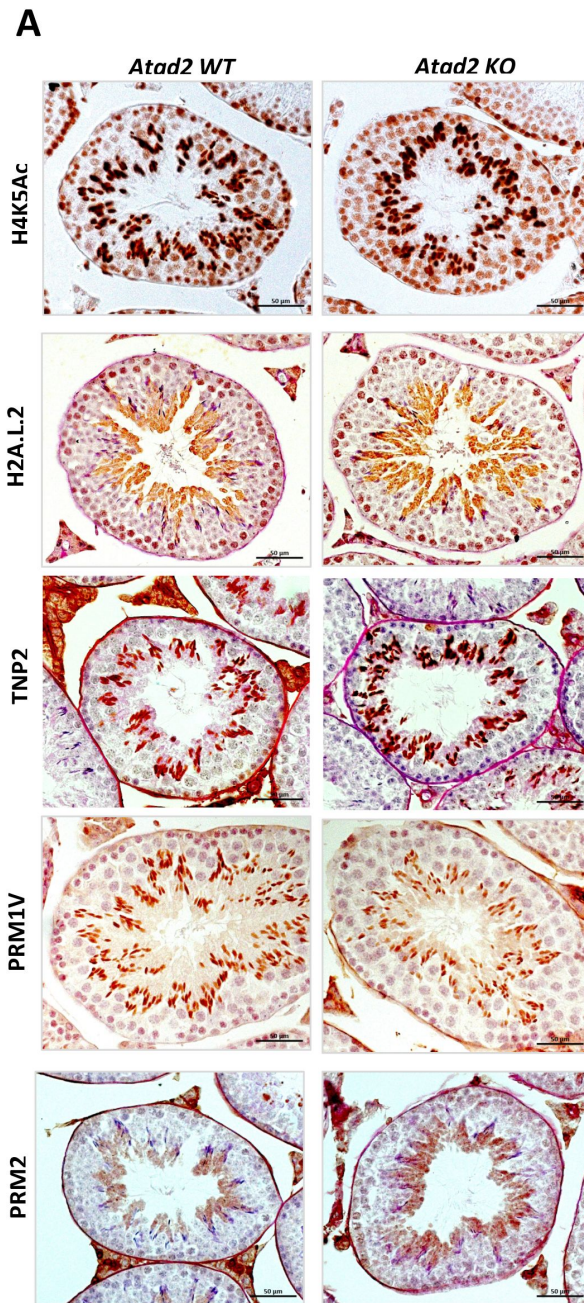


Figure S1

***Atad2* gene inactivation does not significantly disrupt testis histology, the occurrence of H4K5ac, or the cell-specific accumulation of H2A.L.2, TP2, PRM1 and PRM2.**

Sections of paraffin-embedded testes from wild-type or *Atad2* KO mice were used to visualize the accumulation H4K5ac or expression of H2A.L.2, TP2, PRM1 and PRM2.

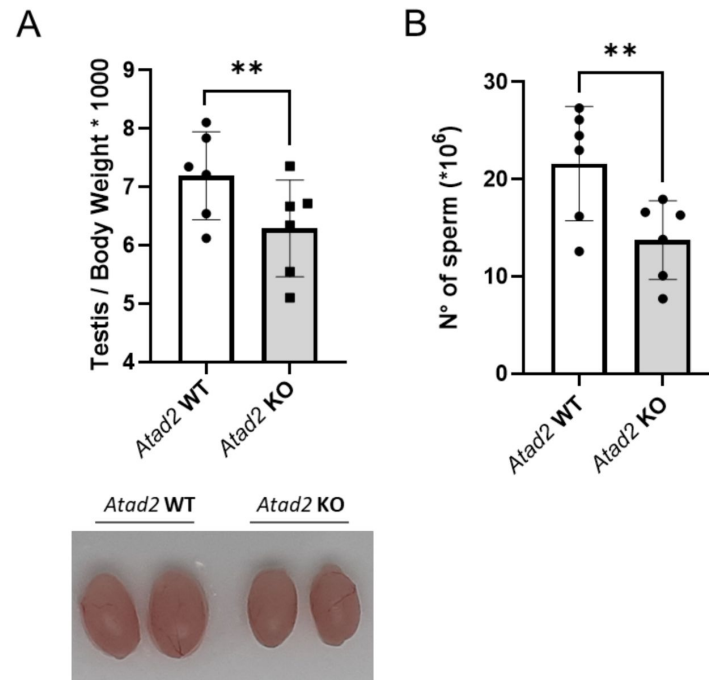


Figure S2

***Atad2* depletion affects testis weight and mature sperm counts.**

(A) Testes from six 21- to 27-week-old wild-type and *Atad2* KO mice were collected, weighted (mg) and their weights normalized to body weight and only same-age mice were compared pairwise. Histograms represent the mean of the measured weights and error bars the Standard Deviation. The *p*-value of pairwise Student t-test = 0.004 (upper panel). The lower panel shows representative testes from wild-type and *Atad2* KO mice. (B) Spermatozoa counts (Cells /ml) were performed for pairs of wild type and *Atad2* KO (*n*=6). The average numbers of spermatozoa counts are respectively 21.6 ± 5.8 and 13.7 ± 4., and the *p*-value of paired Student t-test = 0.001.

Name	Company	Reference	Host Species	Dilution
Anti-ATAD2	Homemade	-	Rabbit	1:1000
anti-H2A.L1/L2 SAB771	Homemade	-	Rabbit	1:200 for IHC
Anti-Histone H3 antibody - Nuclear Marker and ChIP Grade	Abcam	Ab1791	Rabbit	1:5000 for WB
Anti-H2B (testis specific)	Abcam	AB178426	Rabbit	1:100 for IF
anti-H3F3B	Abnova	#H00003021-M01	Mouse	1:200 for IF
Anti-H4K5Ac	PTM Biolabs	#PTM-163	Mouse	1:200 for IHC
Anti-HIRA	Cell Signaling Technology	D6O8L	Rabbit	1:1000 for WB
Pre-PRM2	Homemade	-	Rabbit	1:100 for IF
anti-Protamine 1	BRIAR Patch Biosciences	HUP1N	Mouse	1:200 for IHC/IF
anti-Protamine 2	BRIAR Patch Biosciences	HUP2B	Mouse	1:200 for IHC
Anti-TNP2 (B-2)	Santa Cruz Biotechnology	sc-393843	Mouse	1:200 for IHC
Goat anti Rabbit IgG (H + L)-HRP	Bio-rad	Cat# 1706515	-	1:5000 for WB
Goat anti-Mouse IgG (H + L)-HRP	Bio-rad	Cat# 1706516	-	1:5000 for WB
Alexa Fluor™ 568 goat Anti-mouse IgG (H+L)	Invitrogen	Cat# A11004	-	1:500 for IF
Alexa Fluor™ 488 goat Anti-rabbit IgG (H+L)	Invitrogen	Cat# A11034	-	1:500 for IF

WB: Western Blot Analysis, IF: Immunofluorescence and IHC: Immunohistochemistry.

Table S1

Primary Antibodies

Products	Company	Reference
Bovine Serum Albumin (BSA)	Euromedex	# 04-100-812-C
Clarity™ Western ECL Substrate	Bio-Rad	Cat # 170-5061
Clarity Max™ Western ECL Substrate	Bio-Rad	Cat # 170-5062
Complete Protease inhibitor EDTA-free	Sigma-Aldrich	# 4693159001
Dako Fluorescent Mounting Medium	Dako	S3023
DL-Dithiothreitol (DTT)	Euromedex	EU0006
DNase I recombinant	Roche	4536282001
DPBS	Gibco	14200-075
Ethylendiaminetetraacetic acid disodium salt dihydrate (EDTA)	Sigma-Aldrich	E6635
Ethylene glycol-bis (2-aminoethylether)-N,N,N',N'-tetraacetic acid (EGTA)	Sigma-Aldrich	E3889
Formalin solution, neutral buffered, 10%	Sigma-Aldrich	HT5012
Glutaraldehyde solution	Sigma-Aldrich	G5882
Heparin sodium salt from porcine intestinal mucosa	Sigma-Aldrich	H3149
Hoechst	Invitrogen	# 33342
Human Chorionic Gonadotropin (HCG)	MED-VET	Chorulon 1500 UI
Laemmli Sample Buffer 4x	BioRad	# 1610747
Magnesium Chloride (MgCl₂)	Euromedex	2189
Mayer's Hematoxylin Solution	Sigma-Aldrich	MHS16
M2 medium	Sigma-Aldrich	M7167
M16 medium	Sigma-Aldrich	M7292
Nitrocellulose Blotting membrane	Amersham™ Protran®	10600015
Nonidet™ P 40 Substitute (NP 40)	Sigma-Aldrich	74385
NuPAGE™ 4-12% Bis-Tris Gel	Invitrogen	NP0335BOX

Table S2

Reagents / Products used

Potassium Ferricyanide	Sigma-Aldrich	P-8131
Potassium Ferrocyanide	Sigma-Aldrich	P-9387
Pregnant Mare Serum Gonadotropin (PMSG)	MED-VET	Syncro-part PMSG 600 UI
Saccharose	Sigma-Aldrich	S7903
Saponin from quillaja bark	Sigma-Aldrich	S7900
Sodium deoxycholate (NaDoc)	Sigma-Aldrich	D6750
Superfrost® slides	Thermo Scientific™	LCSF05
TMB ELISA Substrate (High Sensitivity)	Abcam	AB171523
Trans-Blot Turbo 5X Transfer Buffer	BioRad	Cat # 10026938
Triton™ X-100	Sigma-Aldrich	T9284
TRIzol™	Thermo Scientific™	# 15596026
TWEEN® 20	Sigma-Aldrich	P7949
Urea	Sigma-Aldrich	U5378
X-Gal	Invitrogen	B1690
96 Well Round (U) Bottom Plate	Thermo Scientific™	# 163320
450 nm Stop Solution for TMB Substrate	Abcam	AB171529

Table S2 (continued)

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Additional information

Manuscript preparation

Parts of the text have been refined, with the help of free online tools (ChatGPT and DeepL).

Author contributions

AL: experimentation, data interpretation, writing and preparation of the manuscript, FB: supervision, experimentation, data interpretation, writing and preparation of the manuscript, management of acquisition of *Atad2* KO mouse model. SB: experimentation and the maintain of *Atad2* mouse model, EL and CA *in vitro* fertilization essays. FC and EBF transcriptomic data analyses. DePu and CG: transcriptomic data generation, SR: founding acquisition and transcriptomic analysis. DaPe and AV: experimentation and data generation and analyses. SK: conception, founding acquisition and supervision, and coordination of the whole project and writing of the manuscript.

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Reviewer #1 (Public review):

Summary:

The authors analyzed the expression of ATAD2 protein in post-meiotic stages and characterized the localization of various testis-specific proteins in the testis of the Atad2 knockout (KO). By cytological analysis as well as the ATAC sequencing, the study showed that increased levels of HIRA histone chaperone, accumulation of histone H3.3 on post-meiotic nuclei, defective chromatin accessibility and also delayed deposition of protamines. Sperm from the Atad2 KO mice reduces the success of in vitro fertilization. The work was performed well, and most of the results are convincing. However, this manuscript does not suggest a molecular mechanism for how ATAD2 promotes the formation of testis-specific chromatin.

Strengths:

The paper describes the role of ATAD2 AAA+ ATPase in the proper localization of sperm-specific chromatin proteins such as protamine, suggesting the importance of the DNA replication-independent histone exchanges with the HIRA-histone H3.3 axis.

Weaknesses:

(1) Some results lack quantification.

(2) The work was performed well, and most of the results are convincing. However, this manuscript does not suggest a molecular mechanism for how ATAD2 promotes the formation of testis-specific chromatin.

<https://doi.org/10.7554/eLife.107582.1.sa3>

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This manuscript by Liakopoulou et al. presents a comprehensive investigation into the role of ATAD2 in regulating chromatin dynamics during spermatogenesis. The authors elegantly demonstrate that ATAD2, via its control of histone chaperone HIRA turnover, ensures proper H3.3 localization, chromatin accessibility, and histone-to-protamine transition in post-meiotic male germ cells. Using a new well-characterized Atad2 KO mouse model, they show that ATAD2 deficiency disrupts HIRA dynamics, leading to aberrant H3.3 deposition, impaired transcriptional regulation, delayed protamine assembly, and defective sperm genome compaction. The study bridges ATAD2's conserved functions in embryonic stem cells and cancer to spermatogenesis, revealing a novel layer of epigenetic regulation critical for male fertility.

Strengths:

The MS first demonstration of ATAD2's essential role in spermatogenesis, linking its expression in haploid spermatids to histone chaperone regulation by connecting ATAD2-dependent chromatin dynamics to gene accessibility (ATAC-seq), H3.3-mediated transcription, and histone eviction. Interestingly and surprisingly, sperm chromatin defects in Atad2 KO mice impair only in vitro fertilization but not natural fertility, suggesting unknown compensatory mechanisms in vivo.

Weaknesses: The MS is robust and there are not big weaknesses

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Reviewer #3 (Public review):

Summary:

The authors generated knockout mice for *Atad2*, a conserved bromodomain-containing factor expressed during spermatogenesis. In *Atad2* KO mice, HIRA, a chaperone for histone variant H3.3, was upregulated in round spermatids, accompanied by an apparent increase in H3.3 levels. Furthermore, the sequential incorporation and removal of TH2B and PRM1 during spermiogenesis were partially disrupted in the absence of ATAD2, possibly due to delayed histone removal. Despite these abnormalities, *Atad2* KO male mice were able to produce offspring normally.

Strengths:

The manuscript addresses the biological role of ATAD2 in spermatogenesis using a knockout mouse model, providing a valuable *in vivo* framework to study chromatin regulation during male germ cell development. The observed redistribution of H3.3 in round spermatids is clearly presented and suggests a previously unappreciated role of ATAD2 in histone variant dynamics. The authors also document defects in the sequential incorporation and removal of TH2B and PRM1 during spermiogenesis, providing phenotypic insight into chromatin transitions in late spermatogenic stages. Overall, the study presents a solid foundation for further mechanistic investigation into ATAD2 function.

Weaknesses:

While the manuscript reports the gross phenotype of *Atad2* KO mice, the findings remain largely superficial and do not convincingly demonstrate how ATAD2 deficiency affects chromatin dynamics. Moreover, the phenotype appears too mild to elucidate the functional significance of ATAD2 during spermatogenesis.

(1) Figures 4-5: The analyses of differential gene expression and chromatin organization should be more comprehensive. First, Venn diagrams comparing the sets of significantly differentially expressed genes between this study and previous work should be shown for each developmental stage. Second, given the established role of H3.3 in MSCI, the effect of *Atad2* knockout on sex chromosome gene expression should be analyzed. Third, integrated analysis of RNA-seq and ATAC-seq data is needed to evaluate how ATAD2 loss affects gene expression. Finally, H3.3 ChIP-seq should be performed to directly assess changes in H3.3 distribution following *Atad2* knockout.

(2) Figure 3: The altered distribution of H3.3 is compelling. This raises the possibility that histone marks associated with H3.3 may also be affected, although this has not been investigated. It would therefore be important to examine the distribution of histone modifications typically associated with H3.3. If any alterations are observed, ChIP-seq analyses should be performed to explore them further.

(3) Figure 7: While the authors suggest that pre-PRM2 processing is impaired in *Atad2* KO, no direct evidence is provided. It is essential to conduct acid-urea polyacrylamide gel electrophoresis (AU-PAGE) followed by western blotting, or a comparable experiment, to substantiate this claim.

(4) HIRA and ATAD2: Does the upregulation of HIRA fully account for the phenotypes observed in *Atad2* KO? If so, would overexpression of HIRA alone be sufficient to phenocopy the *Atad2* KO phenotype? Alternatively, would partial reduction of HIRA (e.g., through heterozygous deletion) in the *Atad2* KO background be sufficient to rescue the phenotype?

(5) The mechanism by which ATAD2 regulates HIRA turnover on chromatin and the deposition of H3.3 remains unclear from the manuscript and warrants further investigation.

Author response:

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The authors analyzed the expression of ATAD2 protein in post-meiotic stages and characterized the localization of various testis-specific proteins in the testis of the Atad2 knockout (KO). By cytological analysis as well as the ATAC sequencing, the study showed that increased levels of HIRA histone chaperone, accumulation of histone H3.3 on post-meiotic nuclei, defective chromatin accessibility and also delayed deposition of protamines. Sperm from the Atad2 KO mice reduces the success of in vitro fertilization. The work was performed well, and most of the results are convincing. However, this manuscript does not suggest a molecular mechanism for how ATAD2 promotes the formation of testis-specific chromatin.

We would like to take this opportunity to highlight that the present study builds on our previously published work, which examined the function of ATAD2 in both yeast *S. pombe* and mouse embryonic stem (ES) cells (Wang et al., 2021). In yeast, using genetic analysis we showed that inactivation of HIRA rescues defective cell growth caused by the absence of ATAD2. This rescue could also be achieved by reducing histone dosage, indicating that the toxicity depends on histone over-dosage, and that HIRA toxicity, in the absence of ATAD2, is linked to this imbalance.

Furthermore, HIRA ChIP-seq performed in mouse ES cells revealed increased nucleosome-bound HIRA, particularly around transcription start sites (TSS) of active genes, along with the appearance of HIRA-bound nucleosomes within normally nucleosome-free regions (NFRs). These findings pointed to ATAD2 as a major factor responsible for unloading HIRA from nucleosomes. This unloading function may also apply to other histone chaperones, such as FACT (see Wang et al., 2021, Fig. 4C).

In the present study, our investigations converge on the same ATAD2 function in the context of a physiologically integrated mammalian system—spermatogenesis. Indeed, in the absence of ATAD2, we observed H3.3 accumulation and enhanced H3.3-mediated gene expression. Consistent with this functional model of ATAD2— unloading chaperones from histone- and non-histone-bound chromatin—we also observed defects in histone-toprotamine replacement.

Together, the results presented here and in Wang et al. (2021) reveal an underappreciated regulatory layer of histone chaperone activity. Previously, histone chaperones were primarily understood as factors that load histones. Our findings demonstrate that we must also consider a previously unrecognized regulatory mechanism that controls assembled histone-bound chaperones. This key point was clearly captured and emphasized by Reviewer #2 (see below).

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The paper describes the role of ATAD2 AAA+ ATPase in the proper localization of sperm-specific chromatin proteins such as protamine, suggesting the importance of the DNA replication-independent histone exchanges with the HIRA-histone H3.3 axis.

Weaknesses:

(1) Some results lack quantification.

We will consider all the data and add appropriate quantifications where necessary.

(2) The work was performed well, and most of the results are convincing. However, this manuscript does not suggest a molecular mechanism for how ATAD2 promotes the formation of testis-specific chromatin.

Please see our comments above.

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incorporation and removal of TH2B and PRM1 during spermiogenesis, providing phenotypic insight into chromatin transitions in late spermatogenic stages. Overall, the study presents a solid foundation for further mechanistic investigation into ATAD2 function.

Weaknesses:

While the manuscript reports the gross phenotype of Atad2 KO mice, the findings remain largely superficial and do not convincingly demonstrate how ATAD2 deficiency affects chromatin dynamics. Moreover, the phenotype appears too mild to elucidate the functional significance of ATAD2 during spermatogenesis.

We respectfully disagree with the statement that our findings are largely superficial. Based on our investigations of this factor over the years, it has become evident that ATAD2 functions as an auxiliary factor that facilitates mechanisms controlling chromatin dynamics (see, for example, Morozumi et al., 2015). These mechanisms can still occur in the absence of ATAD2, but with reduced efficiency, which explains the mild phenotype we observed.

This function, while not essential, is nonetheless an integral part of the cell's molecular biology and should be studied and brought to the attention of the broader biological community, just as we study essential factors. Unfortunately, the field has tended to focus primarily on core functional actors, often overlooking auxiliary factors. As a result, our decade-long investigations into the subtle yet important roles of ATAD2 have repeatedly been met with skepticism regarding its functional significance, which has in turn influenced editorial decisions.

We chose eLife as the venue for this work specifically to avoid such editorial barriers and to emphasize that facilitators of essential functions do exist. They deserve to be investigated, and the underlying molecular regulatory mechanisms must be understood.

(1) Figures 4-5: The analyses of differential gene expression and chromatin organization should be more comprehensive. First, Venn diagrams comparing the sets of significantly differentially expressed genes between this study and previous work should be shown for each developmental stage. Second, given the established role of H3.3 in MSCI, the effect of Atad2 knockout on sex chromosome gene expression should be analyzed. Third, integrated analysis of RNA-seq and ATAC-seq data is needed to evaluate how ATAD2 loss affects gene expression. Finally, H3.3 ChIP-seq should be performed to directly assess changes in H3.3 distribution following Atad2 knockout.

(1) In the revised version, we will include Venn diagrams to illustrate the overlap in significantly differentially expressed genes between this study and previous work. However, we believe that the GSEAs presented here provide stronger evidence, as they indicate the statistical significance of this overlap (p-values). In our case, we observed p-value < 0.01 (**) and p < 0.001 (***).

(2) Sex chromosome gene expression was analyzed and is presented in Fig. 5C.

(3) The effect of ATAD2 loss on gene expression is shown in Fig. 4A, B, and C as histograms, with statistical significance indicated in the middle panels.

(4) Although mapping H3.3 incorporation across the genome in wild-type and Atad2 KO cells would have been informative, the available anti-H3.3 antibody did not work for ChIP-seq, at least in our hands. The authors of Fontaine et al., 2022, who studied H3.3 during spermatogenesis in mice, must have encountered the same problem, since they tagged the endogenous H3.3 gene to perform their ChIP experiments.

(2) Figure 3: The altered distribution of H3.3 is compelling. This raises the possibility that histone marks associated with H3.3 may also be affected, although this has not been investigated. It would therefore be important to examine the distribution of histone modifications typically associated with H3.3. If any alterations are observed, ChIP-seq analyses should be performed to explore them further.

Based on our understanding of ATAD2's function—specifically its role in releasing chromatin-bound HIRA—in the absence of ATAD2 the residence time of both HIRA and H3.3 on chromatin increases. This results in the detection of H3.3 not only on sex chromosomes but across the genome. Our data provide clear evidence of this phenomenon. The reviewer is correct in suggesting that the accumulated H3.3 would carry H3.3-associated histone PTMs; however, we are unsure what additional insights could be gained by further demonstrating this point.

(3) Figure 7: While the authors suggest that pre-PRM2 processing is impaired in Atad2 KO, no direct evidence is provided. It is essential to conduct acid-urea polyacrylamide gel electrophoresis (AU-PAGE) followed by western blotting, or a comparable experiment, to substantiate this claim.

Figure 7 does not suggest that pre-PRM2 processing is affected in Atad2 KO; rather, this figure—particularly Fig. 7B—specifically demonstrates that pre-PRM2 processing is impaired, as shown using an antibody that recognizes the processed portion of pre-PRM2. ELISA was used to provide a more quantitative assessment; however, in the revised manuscript we will also include a western blot image.

(4) HIRA and ATAD2: Does the upregulation of HIRA fully account for the phenotypes observed in Atad2 KO? If so, would overexpression of HIRA alone be sufficient to phenocopy the Atad2 KO phenotype? Alternatively, would partial reduction of HIRA (e.g., through heterozygous deletion) in the Atad2 KO background be sufficient to rescue the phenotype?

These are interesting experiments that require the creation of appropriate mouse models, which are not currently available.

(5) The mechanism by which ATAD2 regulates HIRA turnover on chromatin and the deposition of H3.3 remains unclear from the manuscript and warrants further investigation.

The Reviewer is absolutely correct. In addition to the points addressed in response to Reviewer #1's general comments (see above), it would indeed have been very interesting to test the segregase activity of ATAD2 (likely driven by its AAA ATPase activity) through in vitro experiments using the *Xenopus* egg extract system described by Tagami et al., 2004. This system can be applied both in the presence and absence (via immunodepletion) of ATAD2 and would also allow the use of ATAD2 mutants, particularly those with inactive AAA ATPase or bromodomains. However, such experiments go well beyond the scope of this study, which focuses on the role of ATAD2 in chromatin dynamics during spermatogenesis.

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